CODE OF BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/booth/booth_mul.vhd [VHDL]

```
1  ----------------------------------------------------------------------------------
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date: 09/10/2026 04:26:32 PM
6  -- Design Name:
7  -- Module Name: booth_mul - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 ----------------------------------------------------------------------------------
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28 --library UNISIM;
29 --use UNISIM.VComponents.all;
30
31 entity booth_mul is
32     Port ( a : in signed (3 downto 0);
33            b : in signed (3 downto 0);
34            p : out signed (7 downto 0));
35 end booth_mul;
36
37 architecture Behavioral of booth_mul is
38 begin
39
40     process (a , b)
41         variable q_pair : std_logic_vector(1 downto 0);
42         variable acc : signed (3 downto 0);
43         variable q : signed (3 downto 0);
44         variable total : std_logic_vector (8 downto 0);
45
```

```
        variable w : integer := 3;

    begin
        q_pair := "00";
        acc := "0000";
        q := b;
        total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);

        for i in 0 to w loop

            q_pair := total(1 downto 0);

            case(q_pair) is
                when "01" => acc := acc + a;
                when "10" => acc := acc -a ;
                when others => null ;
            end case;

            total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);
            total := total(8) & total(8 downto 1);
            acc := signed(total(8 downto 5));
            q := signed(total(4 downto 1));

        end loop;

        p <= signed(total(8 downto 1));

    end process;

end Behavioral;
```

CODE OF TEST BENCH FOR BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/booth/booth_mul_tb.vhd [VHDL]

```
----------------------------------------------------------------------------------
-- Company:
-- Engineer:
--
-- Create Date: 09/10/2026 04:26:32 PM
-- Design Name:
-- Module Name: booth_mul_tb - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
```

```
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity booth_mul_tb is
--  Port ( );
end booth_mul_tb;

architecture testing of booth_mul_tb is


    component booth_mul is
        Port ( a : in signed (3 downto 0);
            b : in signed (3 downto 0);
            p : out signed (7 downto 0));
    end component;


    signal a : signed(3 downto 0);
    signal b : signed(3 downto 0);
    signal p : signed(7 downto 0);

begin


    booth_multiplier_test : booth_mul port map(
        a => a,
        b => b,
        p =>p
    );


    process

    begin
```

```
65          a <= "0101";
66          b <= "1010";
67
68          wait for 10ns;
69
70          a <= "0011";
71          b <= "1100";
72
73          wait for 10ns;
74
75          a <= "1010";
76          b <= "1100";
77
78          wait for 10ns;
79
80          a <= "0001";
81          b <= "1000";
82
83          wait for 10ns;
84
85          a <= "1110";
86          b <= "1011";
87
88          wait for 10ns;
89
90          a <= "1111";
91          b <= "1111";
92
93          wait for 10ns;
94
95          a <= "0000";
96          b <= "0000";
97
98          wait for 10ns;
99
100         a <= "1010";
101         b <= "1111";
102
103         wait for 10ns;
104
105         a <= "0001";
106         b <= "1111";
107
108         wait for 10ns;
109
110         a <= "1010";
111         b <= "1111";
112
113
114         wait;
```

```
115
116     end process;
117
118
119
120 end testing;
```

SCREENSHOT OF CODE

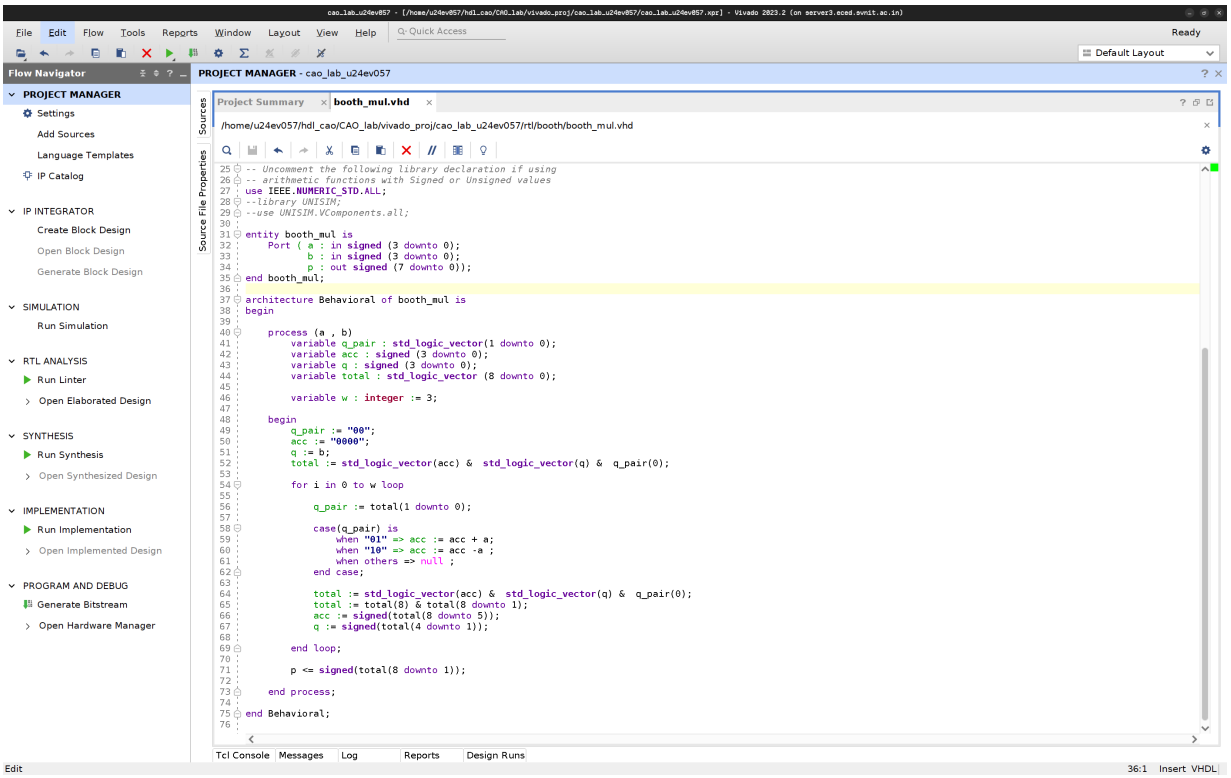


Figure 1: Design

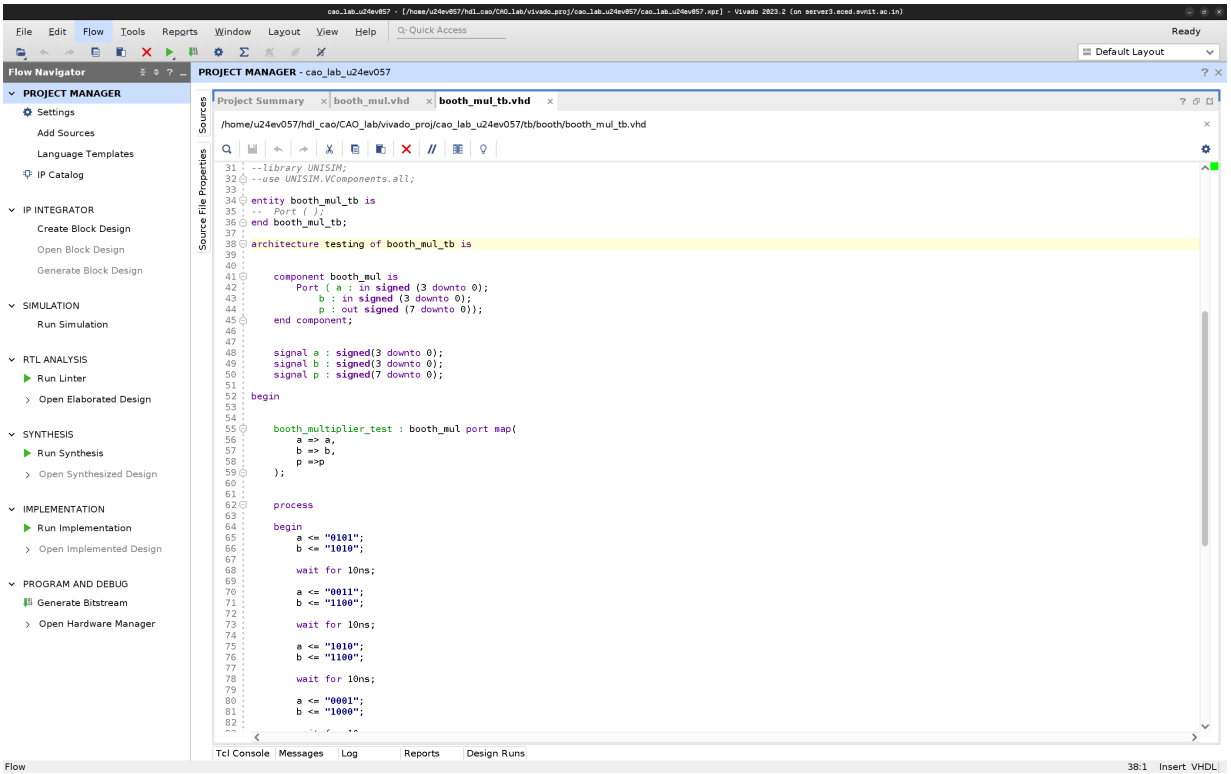


Figure 2: Tb

WAVEFORM

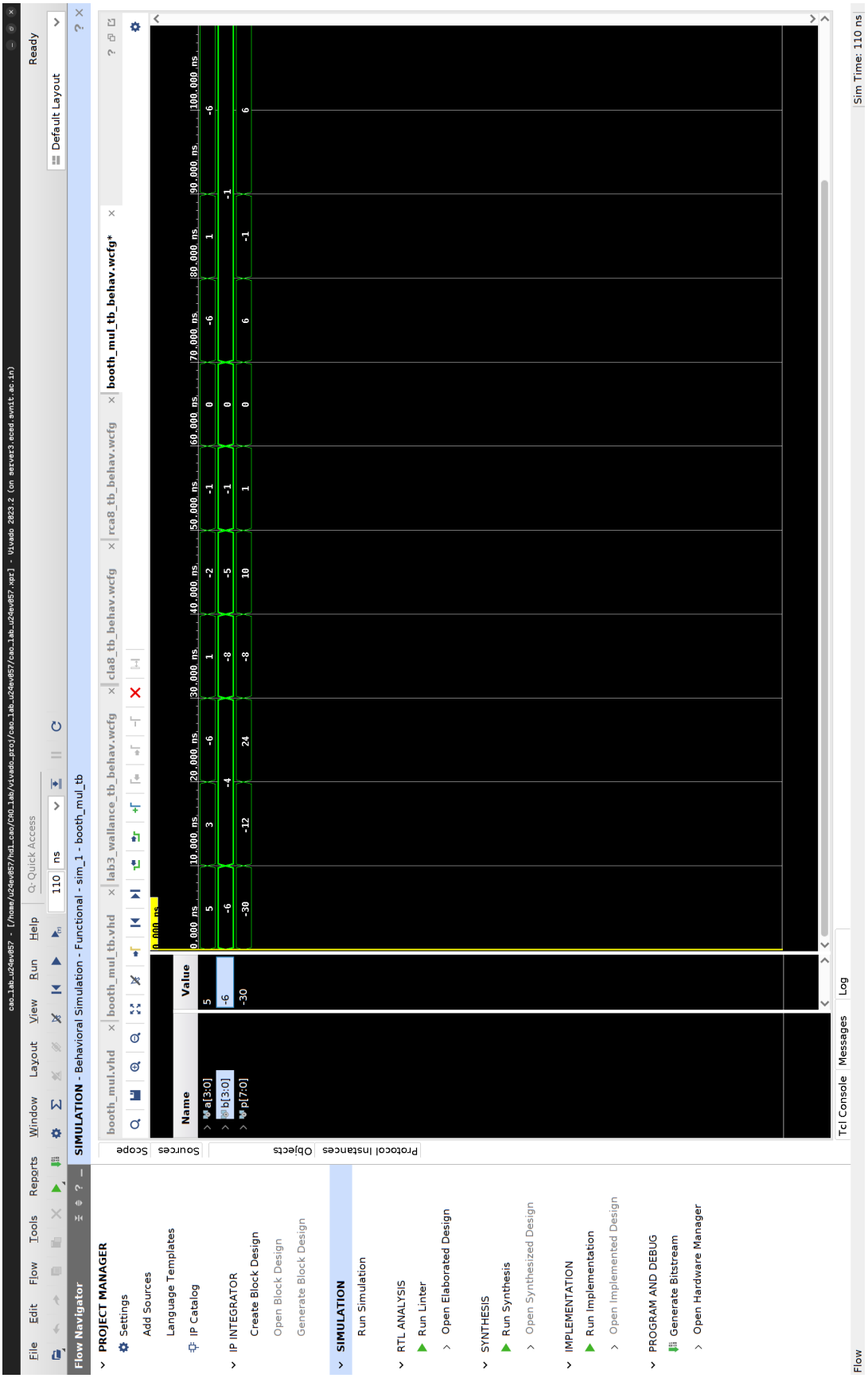


Figure 3: Waveform